

which emphasises extensibility and standard compliance. It implements a majority of the SQL2011 standards and comes with various enterprise class features.

The standard set of features that are expected in any industry DBMS are available in PostgreSQL as well. Here is the list of a few of its features that are gaining popularity:

1. Runs on all major operating systems
 2. Fully ACID compliant for transaction reliability
 3. Multi-version concurrency control to provide immunity to dirty reads and serialisability
 4. Support for SQL data types
 5. Streaming replication to continuously ship and apply Write Ahead Log records to the standby server...
- ...and many more.

Cassandra

Next in the list is Cassandra, ranked the second runner-up by DB-Engines for 2015. It was initially developed by Facebook engineers to power the social media giant's inbox search and was then made open source in 2008. Cassandra is a distributed DBMS specially designed to handle large amounts of data spread across a vast cluster of machines. It offers support for data spanning across multiple data centres with asynchronous masterless replication.

Technically, Cassandra also broke away from the traditional RDBMS 'table and columns' model, with data being stored on a single machine to scale across thousands of servers. Cassandra also places a high value on performance and scalability. A study done by the University of Toronto in 2012 found that among NoSQL databases, Cassandra was the clear winner in terms of scalability, and achieved the highest throughput across the maximum number of nodes.

Even though its creators at Facebook have abandoned Cassandra, it is still providing leading Web infrastructure companies like Twitter, Netflix and Apple. A study done by an Australian company, Solid IT, reveals that Cassandra is the second most popular NoSQL database after MongoDB and the third fastest growing database, over all.

Here is a small list of features that Cassandra offers.

1. **Decentralisation:** Every node in a cluster has the same role; hence, there's no master, and so no single point of failure.
2. **Multiple data centre replication:** It is designed as a distributed system to support nodes across multiple data centres.
3. **Cassandra Query Language (CQL):** It has introduced a new query language called CQL, an SQL-like alternative.
4. **MapReduce Support:** Cassandra provides integration for Hadoop with MapReduce support.

Again, I have picked only a few key features from Cassandra's vast list, leaving the rest for readers to explore.

MariaDB

Another RDBMS software that is gaining popularity is

MariaDB. It has been forked from MySQL as an alternative to provide a drop-in replacement for MySQL, with better and a more enhanced set of features compared to the equivalent MySQL release. It was started by the founder of MySQL, Michael Widenius, after Oracle acquired Sun Microsystems (which had bought MySQL in 2008) in late 2009, early 2010.

MariaDB was always meant to maintain high compatibility with MySQL, and it ensures drop-in replacement with libraries as well as binary equivalence. Some of the features of MariaDB are listed below.

1. **Open and free:** It is developed by the community under GPL and hence is available at no cost.
2. **Support:** MySQL itself has well established documentation support, which can also be used for MariaDB. Apart from that, there is a large enough MariaDB-specific knowledge base/online community.
3. **Speed:** MariaDB is regarded as one of the fastest databases available, even faster than MySQL.
4. **Functionality:** It provides support for all MySQL features, along with additional new and enhanced features developed by the community.
5. **Ease of use:** As per user feedback, MariaDB is easy to use, especially with features like flexible syntax.

My list of DBMSs is merely a point for discussions to take off on the more than 200 such databases. It's not important which one is chosen and which is not; rather, my attempt has been to give the reader an overview of how much the open source world has evolved in database management. Also, industry adoption of open source databases has been increasing, day by day, moving away from traditional proprietary databases. Hence the mantra for success is: "Go open, go big." 

References

- [1] <http://db-engines.com/en/ranking>
- [2] <https://www.mongodb.org/>
- [3] <https://en.wikipedia.org/wiki/MongoDB>
- [4] <http://searchdatamanagement.techtarget.com/definition/MongoDB>
- [5] <https://en.wikipedia.org/wiki/MySQL>
- [6] <https://www.mysql.com/why-mysql/topreasons.html>
- [7] https://wiki.postgresql.org/wiki/Main_Page
- [8] <http://www.postgresql.org/>
- [9] <https://en.wikipedia.org/wiki/PostgreSQL>
- [10] https://en.wikipedia.org/wiki/Apache_Cassandra
- [11] <http://www.wired.com/2014/08/datastar/>
- [12] <https://en.wikipedia.org/wiki/MariaDB>
- [13] <http://www.zdnet.com/article/open-source-mariadb-a-mysql-fork-challenges-oracle/>

By: Ashish Bunkar

The author is a Linux professional with more than five years' experience in the open source field. He likes to share his interests in SAN and virtualisation technologies.